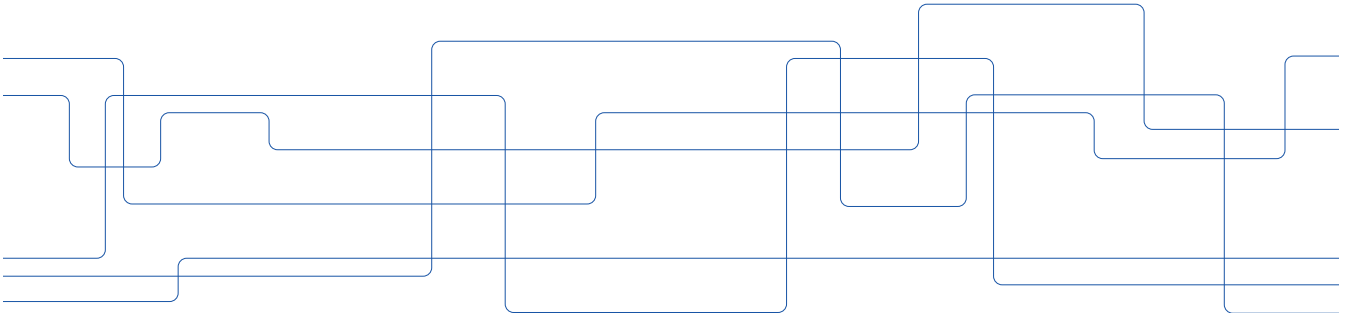


Lecture 3: Data Preprocessing



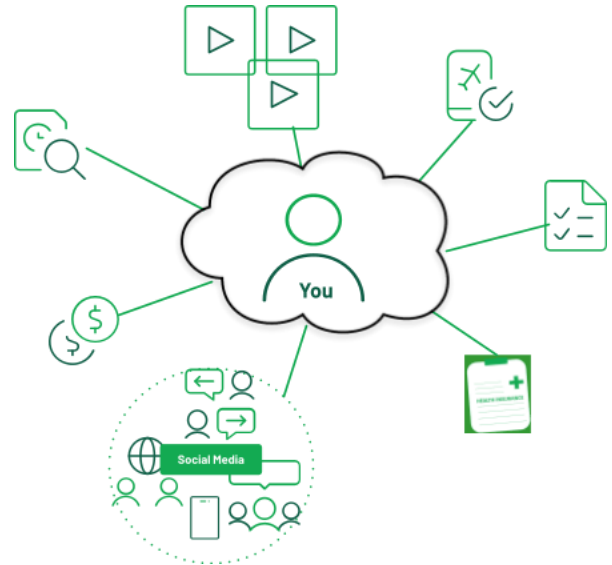
Learning objectives

In previous lectures:

We discussed the importance of data in training AI models

After this lecture, you will be able to:

- Explain the different components of a **modern data ecosystem**
- Explain what data analysis is, and the **key steps** in the data analysis process
- How to Handle **Missing** Data
- Remove or keep **outliers**
- How to **normalize** the data
- How to **visualize** the data efficiently



Importance of Data Analytics

3 Important Statistics About How Much Data Is Created Every Day



1 How much data is generated every minute?

41,666,667

messages shared
by WhatsApp users

1,388,889

video / voice calls made
by people worldwide

404,444

hours of video streamed
by Netflix users

347,222

stories posted by Instagram users

150,000

messages shared by Facebook users

147,000

photos shared by Facebook users

2 Estimated Data Consumption from 2021 to 2024



3 Data Growth in 2021

2 TRILLION

searches on Google by the end of 2021

1.134 TRILLION MB

volume of data created every day

3,026,626

emails sent every second, 67% of which are spam

278,108 PETABYTES

global IP data per month by the end of 2021

230,000

new malware versions created every day

82%

share of video in total global internet
traffic at the end of 2021

*"The constant increase in data processing speeds and bandwidth, the nonstop invention of new tools for creating, sharing, and consuming data, and the steady addition of new data creators and consumers around the world, ensure that data growth continues unabated. **Data begets more data in a constant virtuous cycle.**"*

Forbes 2020 report

Data Analytics:

help organizations extract **value** and **insights** from data, leading to more informed **decision-making** and better **outcomes**

A modern data ecosystem

Data Sources



Enterprise data environment

- Data integrated from disparate sources
- Different types of analysis and skills to generate insights
- Active stakeholders to collaborate and act on insights generated
- Tools, applications, and infrastructure to store, process, and disseminate data as required

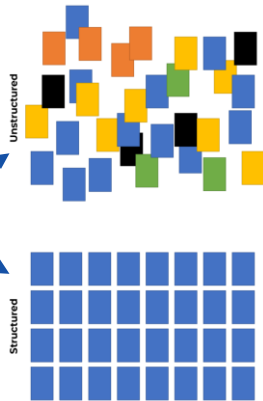
Users



A modern data ecosystem includes a whole network of **interconnected**, **independent**, and **continually evolving** entities.

Data Sources

Data Sources



- Text
- Images
- Videos
- click streams
- user conversations
- social media platforms
- the Internet of things or IoT devices
- real-time events that stream data
- databases
- data sourced from professional data providers and agencies



The sources have never before been so **diverse** and **dynamic**

Different types of Data Analysis

Descriptive Analytics



“What happened?”

- summarizes **past data** and present the findings to stakeholders
- e.g. Identifying the peak loadings

Diagnostic Analytics



“Why did it happen?”

- takes the insights from descriptive analytics to dig deeper to find the **cause** of the outcome.
- e.g. seasonal changes is the reason for peak loadings

Predictive Analytics



“What will happen next?”

- Leverages historical data and trends to **predict future** outcomes
- **Not** to say what will happen in the future, it's objective is to forecast what **might** happen in the future

Prescriptive Analytics



“What should be done about it?”

- Analyzes past decisions and events to suggest which course of **actions** are decided
- e.g. preparing back-up energy technologies
- Self-driving cars are a good example of Prescriptive Analytics.



The Data Analysis Process



1. Understanding of the problem and the desired outcome: Defining where you are and where you want to be



2. Setting a clear metric for evaluating outcomes: What will be measured and how it will be measured?



3. Gathering data: Identify data you require, the sources you need, and best tools



4. Cleaning data: Fixing quality issues in the data and standardizing data from sources



5. Analyzing, and mining data: Extracting, analyzing, and manipulating data from different perspectives to understand trends, and patterns



6. Interpreting results: Evaluating defendability of analysis and circumstances under which may not hold true



7. Communicating the findings: Presenting findings in clear, impactful, and convincing ways that impact decision-making.

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Data cleaning (preprocessing)

1. Missing data
2. Outliers
3. Normalization





Types of Missing Data

Understanding the nature of missing data is critical in determining what treatments can be applied to overcome the lack of data.

1) Missing Completely At Random (MCAR): When missing values are **randomly distributed across all observations**. For example: high school GPA data is missing randomly across all schools in a district.

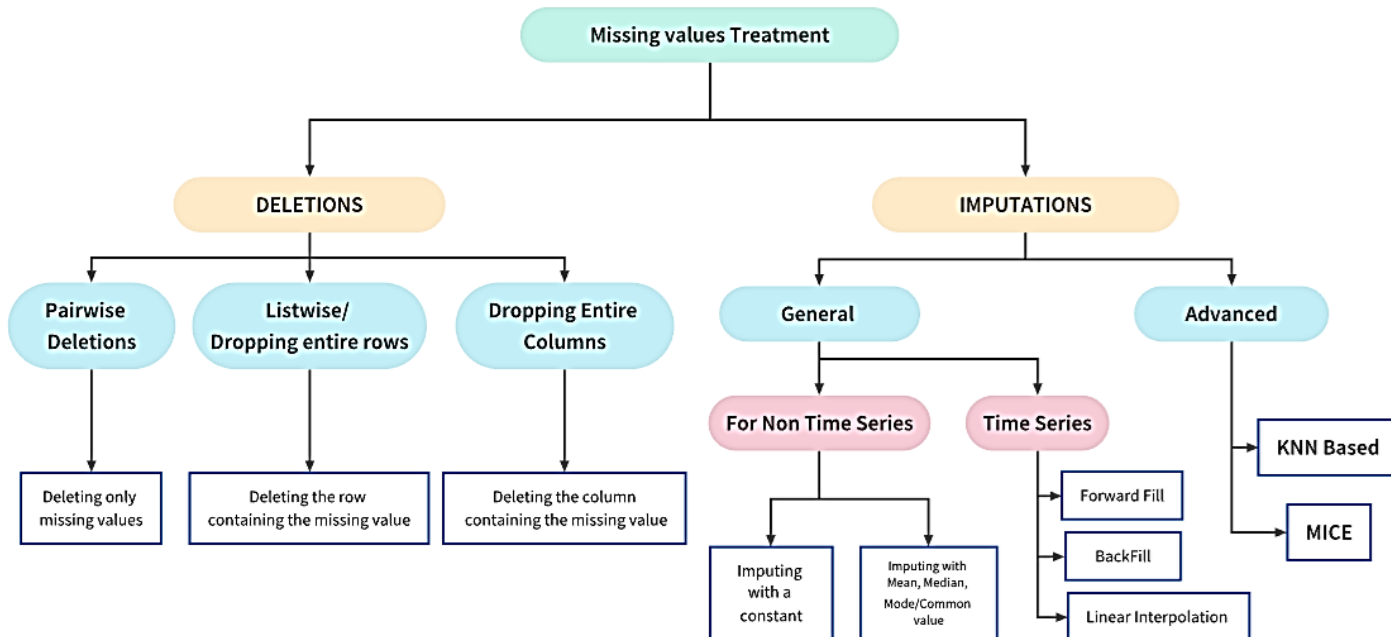
2) Missing At Random (MAR): Missing randomly only within **sub-samples** of data. For example: data is randomly missing for students in specific schools of the district, then the data is MAR.

3) Not Missing At Random (NMAR): When the missing data has a **structure** to it. In the above example, if the data was missing for all students from specific schools, then the data cannot be treated as MAR.

Handling Missing Values

There are 2 primary ways of handling missing values:

1. Deleting the Missing values
2. Modelling (Imputing) the Missing Values



Deleting: listwise

Listwise deletion (complete-case analysis) removes all data for an observation that has one or more missing values.

Particularly if the missing data is limited to a small number of observations, you may just opt to eliminate those cases from the analysis.

It is often disadvantageous to use listwise deletion. This is because the assumptions of MCAR (Missing Completely at Random) are typically rare to support.

As a result, listwise deletion methods produce biased parameters and estimates.

<i>Hour</i>	<i>EV availability profile</i>	<i>Demand (in kWh)</i>	<i>Price (in SEK/kWh)</i>	<i>Solar PV (in kWh)</i>
0	1	3.620145	0.03639	0
1	1	4.1733	0.03654	0
2	1	0.42471	0.03097	0
3	1	5.53704	0.01998	0
4	1		0.0216	0.045
5	1	0.34746	0.0384	0.105
6	1	4.08489	0.04104	0.21
7	1	0.447555	0.04492	0.36
8	0	0.653925	0.04905	0.54
9	0	4.496865	0.05126	0.72
10	0	1.65735	0.05122	0.72

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Deleting: pairwise

Pairwise deletion analyses all cases in which the variables of interest are present and thus maximizes all data available by an analysis basis.

A strength to this technique is that it increases power in your analysis but it has many disadvantages.

It assumes that the missing data are MCAR.

If you delete pairwise then you'll end up with different numbers of observations contributing to different parts of your model, which can make interpretation difficult.

<i>Hour</i>	<i>EV availability profile</i>	<i>Demand (in kWh)</i>	<i>Price (in SEK/kWh)</i>	<i>Solar PV (in kWh)</i>
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Deleting: dropping variables

It is always better to keep data than to discard it.

Sometimes you can drop variables if the data is missing for more than 60% observations but only if that variable is insignificant.

Having said that, imputation is always a preferred choice over dropping variables

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Imputation

Instead of deletion, data scientists have multiple solutions to impute the value of missing data. Depending why the data are missing, imputation methods can deliver reasonably reliable results.

1) Replacing with an arbitrary value

Assigning a fixed value to replace missing values.

This approach is commonly used for **categorical** variables where missing values can be replaced by a new category or by assigning the most common category.

For **numerical** variables, missing values can be replaced by a fixed value such as the **mean**, **median**, or **mode**.

However, this approach can introduce **bias** into the data and should be used with caution as the assigned value may not reflect the true value of the missing data.



Imputation

2-1) Replacing with the mean

- This is the most common method of imputing missing values of numeric columns.
- If there are outliers, then the mean will not be appropriate. In such cases, outliers need to be treated first

2-2) Replacing with the mode

- Mode is the most frequently occurring value.
- It is used in the case of categorical features.

2-3) Replacing with the Median

- The median is the middlemost value.
- It's better to use the median value for imputation in the case of outliers.

Imputation

3-1) Replacing with the previous value – forward fill

In some cases, imputing the values with the previous value instead of the mean, mode, or median is more appropriate.

It is mostly used in time series data.

	date	fruit	price	
Apple Group	0	2021-01-01	apple	0.8
	1	2021-01-02	apple	NaN
	2	2021-01-03	apple	NaN
	3	2021-01-04	apple	1.2
Mango Group	4	2021-01-01	mango	NaN
	5	2021-01-02	mango	3.1
	6	2021-01-03	mango	NaN
	7	2021-01-04	mango	2.8

→

	date	fruit	price
0	2021-01-01	apple	0.8
1	2021-01-02	apple	0.8
2	2021-01-03	apple	0.8
3	2021-01-04	apple	1.2
4	2021-01-01	mango	3.1
5	2021-01-02	mango	3.1
6	2021-01-03	mango	3.1
7	2021-01-04	mango	2.8

3-2) Replacing with the next value – backward fill

In backward fill, the missing value is imputed using the next value.

	date	fruit	price
0	2021-01-01	apple	0.8
1	2021-01-02	apple	NaN
2	2021-01-03	apple	NaN
3	2021-01-04	apple	1.2
4	2021-01-01	mango	NaN
5	2021-01-02	mango	3.1
6	2021-01-03	mango	NaN
7	2021-01-04	mango	2.8

→

	date	fruit	price
0	2021-01-01	apple	0.8
1	2021-01-02	apple	1.2
2	2021-01-03	apple	1.2
3	2021-01-04	apple	1.2
4	2021-01-01	mango	3.1
5	2021-01-02	mango	3.1
6	2021-01-03	mango	2.8
7	2021-01-04	mango	2.8

4) Linear interpolation

Linear interpolation is a commonly used method for handling missing data in a dataset. It involves using the values of neighboring data points to estimate the missing value.

Suppose we have a dataset with a missing value between two data points, x_1 and x_2 . To estimate the missing value, we can use linear interpolation by fitting a straight line between x_1 and x_2 and using the equation of the line to estimate the missing value.

The equation of the line can be expressed as:

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

Linear interpolation assumes that the relationship between the independent and dependent variables is linear and that the missing value falls on this line.

It is a simple and quick method for handling missing data, but it may not always be appropriate if the relationship between the variables is not linear or if there are significant variations in the data.

Imputation

5) KNN (Kth Nearest Neighbors)

KNN is a machine learning algorithm that can be used for handling missing data. In this method, the missing values are estimated by taking into account the values of the k nearest neighbors in the data set.

The advantage of KNN imputation is that it takes into account the relationships between variables and can work well for non-linear data.

However, it can be computationally expensive and sensitive to the choice of k and the similarity measure used. It may also introduce bias if the missingness is related to the value of the variable being imputed.

X	Y
	1
	$? = (1 + 0.5) / 2$
	0.5
	0.1
	$? = (0.1 + 10) / 2$
	10
	0.03



Quiz

What is the missing value?

Ambient temperature ©	wastewater Temp @buildings ©	wastewater Flowrate @ buildings (m3/s)	Rain (m3/s)	Wastewater Temperature @ WWTP (C)
4.90	25.46	1.57	1.43	15.41
5.30	25.50	1.61	1.30	15.29
5.80	25.51	1.60	1.36	14.99
6.30	25.52	1.64	1.36	14.84
6.00	25.38	1.72	1.13	14.52
6.10	25.33	1.59	1.37	14.15
5.80	25.13	1.59	1.75	13.80
5.30	25.08	1.66		13.40
4.70	25.02	1.70	1.27	12.93
4.40	24.73	1.66	1.05	12.60
4.10	24.51	1.66	0.96	12.21
3.90	24.49	1.66	1.38	11.96
3.60	24.36	1.65	1.71	11.93



Quiz

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Ambient temperature ©	wastewater Temp @buildings ©	wastewater Flowrate @ buildings (m3/s)	Rain (m3/s)	Wastewater Temperature @ WWTP (C)
4.90	25.46	1.57	1.43	15.41
5.30	25.50	1.61	1.30	15.29
5.80	25.51	1.60	1.36	14.99
6.30	25.52	1.64	1.36	14.84
6.00	25.38	1.72	1.13	14.52
6.10	25.33	1.59	1.37	14.15
5.80	25.13	1.59	1.75	13.80
5.30	25.08	1.66	1.47	13.40
4.70	25.02	1.70	1.27	12.93
4.40	24.73	1.66	1.05	12.60
4.10	24.51	1.66	0.96	12.21
3.90	24.49	1.66	1.38	11.96
3.60	24.36	1.65	1.71	11.93



Imputation

6) Multiple imputation using chained equations (MICE)

Multiple Imputation by Chained Equations is a robust, informative method of dealing with missing data in datasets.

The procedure 'fills in' (imputes) missing data in a dataset through an iterative series of predictive models. In each iteration, each specified variable in the dataset is imputed using the other variables in the dataset. These iterations should be run until it appears that convergence has been met.

It's an iterative process. Impute X_1 based on $X_2, \dots, X_p$. Then impute X_2 , based on X_1 and $X_3, \dots, X_p$. And continue down the line.

This process is continued until all specified variables have been imputed. Additional iterations can be run if it appears that the average imputed values have not converged.



Imputation

6) MICE (Example)

A bank wants to sell its credit cards.

It reaches out to 6 people and jots down these data samples.

Age	Experience	Salary	Purchased
25		50	0
27	3		1
29	5	110	1
31	7	140	0
33	9	170	1
	11	200	0

Imputation

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A bank wants to sell its credit cards.

It reaches out to 6 people and jots down these data samples.

Age	Experience	Salary	Purchased
25		50	0
27	3		1
29	5	110	1
31	7	140	0
33	9	170	1
	11	200	0

The real values are:

Keep them in your mind for validation later.

Age	Experience	Salary	Purchased
25	1	50	0
27	3	80	1
29	5	110	1
31	7	140	0
33	9	170	1
35	11	200	0

Imputation

6) MICE (Example)

First we get rid of purchased column, because we have all the data.

Age	Experience	Salary
25		50
27	3	
29	5	110
31	7	140
33	9	170
	11	200

Imputation

6) MICE (Example)

First we get rid of purchased column, because we have all the data.

Age	Experience	Salary
25		50
27	3	
29	5	110
31	7	140
33	9	170
	11	200

Iteration 1:

Step 1: Impute all missing values with the mean of their respective columns as an initial guess. We'll call this the "Zeroth" data set.

Age	Experience	Salary
25	7	50
27	3	134
29	5	110
31	7	140
33	9	170
29	11	200

Imputation

6) MICE (Example)

Step 2: Remove the “age” imputed value(s).

Age	Experience	Salary
25	7	50
27	3	134
29	5	110
31	7	140
33	9	170
	11	200

Imputation

6) MICE (Example)

Step 2: Remove the “age” imputed value(s).

Age	Experience	Salary
25	7	50
27	3	134
29	5	110
31	7	140
33	9	170
	11	200

Step 3: The remaining features and rows become the feature matrix, “age” becomes the target variable. We use [Linear Regression](#) model on the fully filled rows to estimate the missing age.

The predicted age is 36.2532

Age	Experience	Salary
25	7	50
27	3	134
29	5	110
31	7	140
33	9	170
	11	200

y

X₁

X₂

Imputation

6) MICE (Example)

Step 2: Remove the “age” imputed value(s).

Age	Experience	Salary
25	7	50
27	3	134
29	5	110
31	7	140
33	9	170
	11	200

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Age	Experience	Salary
25	7	50
27	3	134
29	5	110
31	7	140
33	9	170
36.2532	11	200

y

X₁

X₂

Imputation

6) MICE (Example)

Step 4: Remove the “experience” imputed value(s). The remaining features and rows become the feature matrix, “experience” becomes the target variable, and we run a Linear Regression model on the fully filled rows to estimate the missing experience.

The predicted experience value is 1.8538.

Age	Experience	Salary
25		50
27	3	134
29	5	110
31	7	140
33	9	170
36.2532	11	200

X_1

y

X_2

Imputation

6) MICE (Example)

Step 4: Remove the “experience” imputed value(s). The remaining features and rows become the feature matrix, “experience” becomes the target variable, and we run a Linear Regression model on the fully filled rows to estimate the missing experience.

The predicted experience value is 1.8538.

Age	Experience	Salary
25	1.8538	50
27	3	134
29	5	110
31	7	140
33	9	170
36.2532	11	200

X_1

y

X_2

Imputation

6) MICE (Example)

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Age	Experience	Salary
25	1.8538	50
27	3	134
29	5	110
31	7	140
33	9	170
36.2532	11	200

Step 5: The same process for salary.

The **iteration 1** is finished.

Age	Experience	Salary
25	1.8538	50
27	3	72.7748
29	5	110
31	7	140
33	9	170
36.2532	11	200

Imputation: MICE (Example)

6) **Step 6:** we subtract two latest datasets.

Age	Experience	Salary
25	1.8538	50
27	3	72.7748
29	5	110
31	7	140
33	9	170
36.2532	11	200

First dataset

minus

Age	Experience	Salary
25	7	50
27	3	134
29	5	110
31	7	140
33	9	170
29	11	200

Zeroth dataset

=

Age	Experience	Salary
0	-5.1462	0
0	0	-61.225
0	0	0
0	0	0
0	0	0
7.2532	0	0

Difference

Imputation: MICE (Example)

6) **Step 6:** we subtract two latest datasets.

Age	Experience	Salary
25	1.8538	50
27	3	72.7748
29	5	110
31	7	140
33	9	170
36.2532	11	200

First dataset

minus

Age	Experience	Salary
25	7	50
27	3	134
29	5	110
31	7	140
33	9	170
29	11	200

Zeroth dataset

=

Age	Experience	Salary
0	-5.1462	0
0	0	-61.225
0	0	0
0	0	0
0	0	0
7.2532	0	0

Difference

Now we repeat the steps 2-6 with this new dataset, until we get a stable model, i.e. until the difference between the two “latest” imputed datasets becomes very small.

Step 2 was removing the “age” missing value and estimating it using the Linear Regression.

Imputation: MICE (Example)

6) Step 6: we subtract two latest datasets.

Age	Experience	Salary	minus	Age	Experience	Salary	=	Age	Experience	Salary
25	1.8538	50		25	7	50		0	-5.1462	0
27	3	72.7748		27	3	134		0	0	-61.225
29	5	110		29	5	110		0	0	0
31	7	140		31	7	140		0	0	0
33	9	170		33	9	170		0	0	0
36.2532	11	200		29	11	200		7.2532	0	0
First dataset				Zeroth dataset				Difference		

Now we repeat the steps 2-6 with this new dataset, until we get a stable model, i.e. until the difference between the two “latest” imputed datasets becomes very small.

Step 2 was removing the “age” missing value and estimating it using the Linear Regression.

We stop iterations when a pre-defined threshold is breached.

Or we do it until a pre-defined maximum number of iterations get completed.

Imputation: MICE (Example)

Iteration 2

Age	Experience	Salary
25	1.8538	50
27	3	72.7748
29	5	110
31	7	140
33	9	170
36.2532	11	200

First dataset

minus

Age	Experience	Salary
25	0.9172	50
27	3	80.7385
29	5	110
31	7	140
33	9	170
34.8732	11	200

Second dataset

=

Age	Experience	Salary
0	0.9366	0
0	0	7.9637
0	0	0
0	0	0
0	0	0
1.38	0	0

Difference

Imputation: MICE (Example)

Iteration 2

Age	Experience	Salary
25	1.8538	50
27	3	72.7748
29	5	110
31	7	140
33	9	170
36.2532	11	200

First dataset

minus

Age	Experience	Salary
25	0.9172	50
27	3	80.7385
29	5	110
31	7	140
33	9	170
34.8732	11	200

Second dataset

=

Age	Experience	Salary
0	0.9366	0
0	0	7.9637
0	0	0
0	0	0
0	0	0
1.38	0	0

Difference

Imputation: MICE (Example)

Iteration 4

Age	Experience	Salary
25	1.0015	50
27	3	79.9876
29	5	110
31	7	140
33	9	170
35.0019	11	200

Third dataset

minus

Age	Experience	Salary
25	0.9999	50
27	3	80.0007
29	5	110
31	7	140
33	9	170
34.9998	11	200

Forth dataset

=

Age	Experience	Salary
0	0.0016	0
0	0	0.0131
0	0	0
0	0	0
0	0	0
0.002	0	0

Difference

Advantage: higher accuracy

Disadvantage: Time consuming

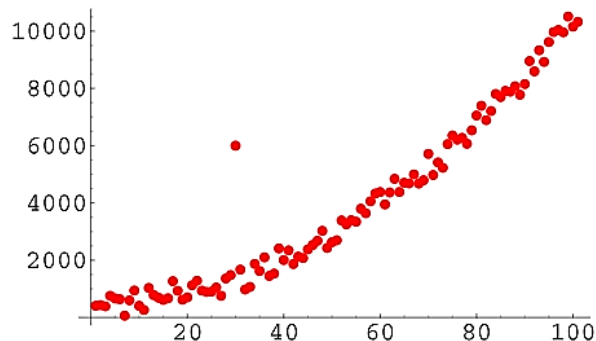
Outliers in Data

An outlier is an observation that lies **abnormally** far away from other values in a dataset.

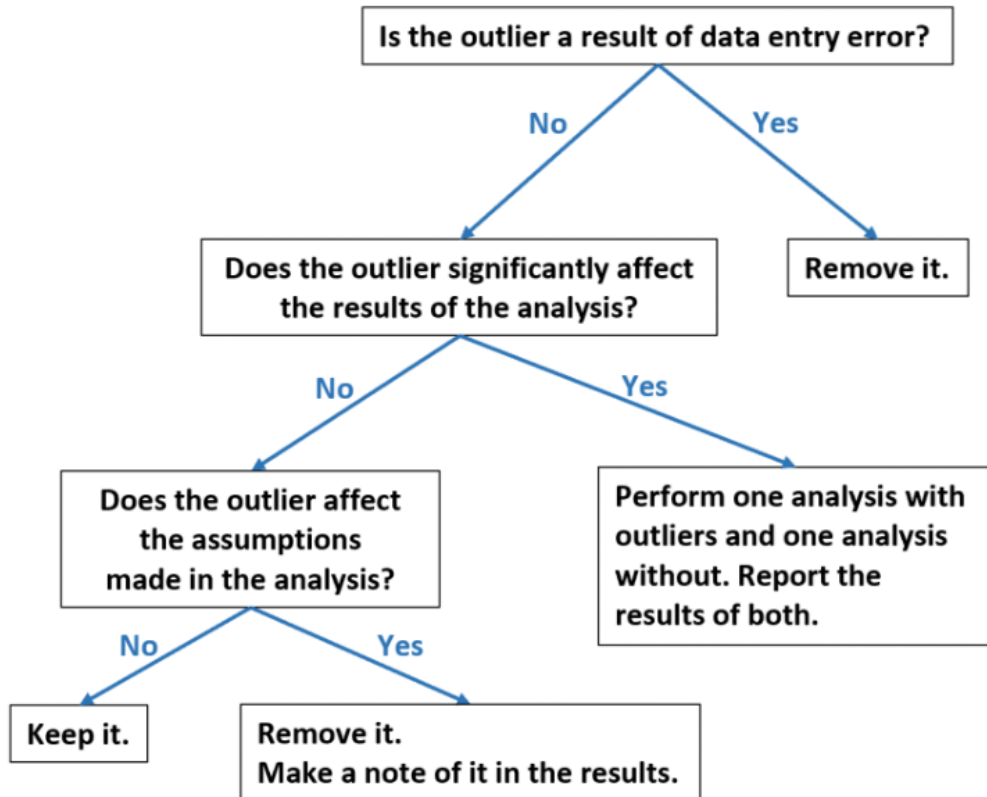
Outliers can be problematic because they can affect the results of an analysis.

However, they can also be informative about the data you're studying because they can reveal abnormal cases or individuals that have rare traits.

In any analysis, you must decide to **remove** or **keep** outliers.



When to Remove Outliers in Data





Is the Outlier a Result of Data Entry Error?

Sometimes outliers in a dataset are simply a result of data entry error.

Suppose a biologist is collecting data on the height of a certain species of plants and records the following data:

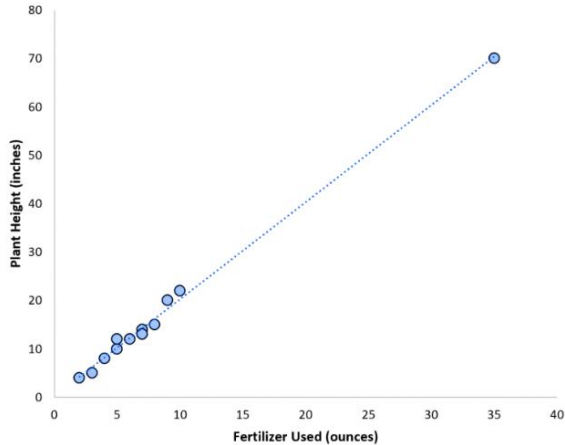
Clearly the entry for 755 inches is an outlier and is likely a result of data entry error.

If the biologist kept this observation and calculated a descriptive statistic like the mean height of the plants in the sample, this observation would greatly skew the results and give an inaccurate picture of the true mean height of the plants.

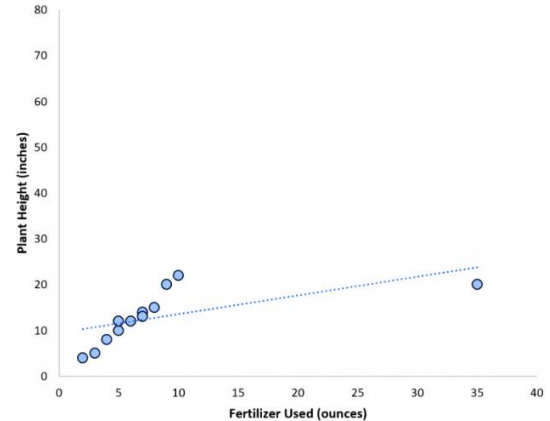
In this scenario (and in scenarios similar to this one) it makes sense to remove this outlier from the dataset because it's an error and is not a legitimate data point to include in the analysis.

- 6.83 inches
- 7.51 inches
- 5.21 inches
- 5.84 inches
- 7.83 inches
- **755 inches**
- 6.53 inches
- 6.31 inches
- 5.91 inches

Does the Outlier Significantly Affect the Results of the Analysis?



In this scenario, the outlier doesn't actually violate any of the assumptions of a linear regression model, so we could keep it in the dataset.



This outlier significantly affects the regression line so we could fit one regression model with the outlier and one without, then report the results of both regression models.



Normalization

Normalization is a process of scaling numeric features of a dataset to a common range, typically between 0 and 1, to avoid biases due to the difference in the scale of features. Here are some common ways of normalizing a dataset:

Min-max normalization:

It scales the dataset to a range between 0 and 1, using the minimum and maximum values of each feature.

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Z-score normalization:

It transforms the dataset so that it has a mean of 0 and a standard deviation of 1.

$$x_{scaled} = \frac{x - mean}{sd}$$

Decimal scaling normalization: It scales the dataset by shifting the decimal point of each feature to the left or right, so that the largest absolute value is less than 1.

$$U_i = \frac{V_i}{10^j}$$



Visualization

What is Data Visualization?

In simple terms, Data Visualization (DataViz) is the process of generating graphical representations of data for various purposes. These graphical representations are commonly known as plots or charts in data science terminology.

Why is Data Visualization Important in Data Science?

There are many reasons for data visualization in data science. Data visualization benefits include

- Communicating your results or findings,
- Monitoring the model's performance at the evaluation stage,
- Hyperparameter tuning,
- Identifying trends, patterns and correlation between dataset features,
- Data cleaning such as outlier detection, and
- Validating model assumptions



What Makes Data Visualization Effective?

To get the most out of data visualization, you should consider the following things. These are the fundamentals of data visualization.

- **Clarity:** Data should be visualized in a way that everyone can understand.
- **Problem domain:** When presenting data, the visualizations should be related to the business problem.
- **Interactivity:** Interactive plots are useful to compare and highlight certain things within the plot.
- **Comparability:** We can compare the things easily with good plots.
- **Aesthetics:** Quality plots are visually aesthetic.
- **Informative:** A good plot summarizes all relevant information.



Tools and Software for Data Visualization

Python provides open-source libraries such as

- Matplotlib
- Seaborn
- Plotty
- Bokeh
- Altair

R provides open-source libraries such as

- Ggplot2
- Lattice

Other data visualization libraries

- IBM SPSS
- Minitab
- Matlab for data visualization
- Tableau

Tableau and Microsoft Power BI are popular among data scientists.

Conclusion

"Visualization gives you answers to questions you didn't know you had."

Ben Schneiderman, American computer scientist and information visualization expert.





Next

Go to CANVAS and do HW0: Data preprocessing



References

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